



Review

How does the manipulation of time pressure during soccer tasks influence physical load and mental fatigue?

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ABSTRACT

This study aimed to analyze whether having more or less time to solve soccer tasks influences physical load and mental fatigue. 48 semi-professional soccer players, from teams in national leagues participated ($M_{age} = 22.4$, $SD_{age} = 2.25$). They carried out 2 sessions with 4 tasks in each session. In one session, there was more time available; in the other, there was less time available. GPS technology was used to measure physical load, and an adaptation of NASA and Visual Analog Scale to measure mental load and mental fatigue. A related-samples T-test and magnitude based on inference were used to determine the possible effect. Soccer players reported that tasks with less available time were more mentally demanding. Moreover, less available time to solve the tasks significantly increased players' RPE and decreased their Heart Rate and external physical load. Therefore, the available time significantly affects mental and physical load and mental fatigue.

1. Introduction

A dynamic framework in soccer involves interrelated physical, technical-tactical, and psychological demands (Balagué et al., 2017). To obtain these specific adaptations in soccer players, tasks simulating competitive-specific scenarios have proliferated, focusing on training goals through constraints (Vilar et al., 2012). However, as knowing the effects of constraints on training load is important, this topic has focused on the physical load (Bujalance-Moreno et al., 2019). In contrast, less is known about the effects of task constraints on mental fatigue (Alarcón et al., 2018). The manipulation of the available time as a constraint has been widely used in soccer training tasks but, to our knowledge, no previous studies have tested its effects on physical load and mental fatigue.

Soccer training and competition load can be mentally and physically fatiguing (Nédélec et al., 2012). Physical load is divided into external load, or the physical effort prescribed (i.e., meters or sets), and internal load (Impellizzeri et al., 2019), or the psychophysiological response produced by each player (i.e., lactate or heart rate). These efforts can lead to soccer-specific physical fatigue (Halsom, 2014), impairing the traditional physiological systems (i.e., lactate or heart rate; Van Cutsem et al., 2020).

On the other hand, mental load, defined as the mental effort needed to solve a soccer task (García-Calvo, 2017), can produce mental fatigue. Mental fatigue is defined as a psychobiological state with multiple derivations caused by prolonged mental load (Van Cutsem et al., 2017). Specific mental fatigue mechanisms have been observed, concluding that this variable is different from physical fatigue (Van Cutsem et al., 2020). During sports events, mental fatigue may be caused by thoughts and emotions (Van Cutsem et al., 2017).

Although fatigue has been mainly analyzed from a physical viewpoint, the importance of analyzing mental fatigue in soccer is supported by its negative influence on soccer performance (Kunrath et al., 2020). Mental fatigue decreases the number of high- and low-intensity running actions because soccer players feel increased physical exertion (Coutinho et al., 2017; Smith et al., 2015). Mentally fatigued players perform fewer successful technical actions (i.e., passes, entries, and controls; Badin et al., 2016) and exhibit poor tactical decisions (i.e., distances between teammates or decision making; Coutinho et al., 2017, 2018; Smith et al., 2016). Based on this information, mental fatigue may impair players' physical and technical-tactical soccer performance. The psychobiological model of Marcora et al. (2009) and the strength model of self-control confirmed by Englert (2019) and Brown et al. (2020)—also for studies with less than 30-min of tasks that imply cognitive

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exertion—may explain why mental fatigue impairs soccer performance. It has been shown that mental fatigue significantly increases the ratio of athletes' perceived exertion and decreases their motivation (Van Cutsem et al., 2017). When athletes start a competition, they can maintain their efforts or performance while motivated and/or use self-control to hold positive thoughts (Englert et al., 2015). Ego-depletion and the psychological model indicate that the accumulation of mental fatigue during a task that implies cognitive exertion may increase adenosine in the brain, which impairs all these mechanisms (i.e., an increase in players' RPE slowly worsens self-control and thoughts) and produces an increase in feelings of fatigue, consequently impairing their physical, technical, and tactical performance.

In sports, mental fatigue can be caused by cognition and emotions (Van et al., 2017). Cárdenas et al. (2015) highlighted the role of entropy in cognitive complexity. It has been shown that changes in the scoring system (García-Calvo et al., 2019), technical-tactical rules during decision-making (García-Calvo et al., 2021), or non-oriented goals (Ponce-Bordón et al., 2021) increased mental fatigue through entropy. Concerning emotions, Beniscelli et al. (2014) highlighted the role of interdependence, and the use of general encouragement has been related to mental fatigue due to the negative emotions perceived by players (Díaz-García et al., 2021). Regarding the available time as a constraint, it has been suggested that time constraints may lead to mental fatigue. As previously mentioned, cognition and emotions may cause mental fatigue. On the one hand, Cardenas et al. (2015) suggested that limited decision-time implies higher cognitive processing because players must make decisions faster. Capraro and Cococcioni (2016) indicate that time-pressure may lead to states of ego-depletion due to an overload of the cognitive processing system, consequently favoring the appearance of automatic responses and performance impairments. On the other hand, negative emotions may appear in the presence of time constraint-related ego depletion because performing well is more difficult (e.g., stress, anxiety, or nervousness). Englert et al. (2015) indicated that time pressure might cause negative emotions, and Cardenas et al. (2015) reported an increase in the feelings of anxiety and stress under time constraints. Therefore, although no previous studies have tested the effects of this constraint on mental fatigue, time constraints may lead to mental fatigue due to cognitive and emotional efforts. The lack of information about the brain mechanisms that cause mental fatigue makes it more difficult to explain. Time constraints may even lead to higher physical efforts as compensation, but this has not yet been tested.

The information about the influence of the available time on fatigue may improve coaches' use of time because time is widely used during soccer training. However, to our knowledge, available time as a constraint has not been analyzed in depth, and more research is necessary. Therefore, this study aimed to determine whether having more or less time to solve soccer tasks influences physical load and mental fatigue. Following Cárdenas et al. (2015), we hypothesized that less available time would imply higher mental fatigue (*Hypothesis 1*) and higher internal (*Hypothesis 2a*) and external (*Hypothesis 2b*) physical load than soccer scenarios with more available time.

2. Methods

2.1. Participants

A priori power analysis, using G*Power software 3.1.9.2 (Faul et al., 2007), was performed to determine the number of participants required (95% power at $p < .05$, and a medium effect size of $d = 0.25$, based on the results reported by Díaz-García et al. (2021) and García-Calvo et al. (2021). The minimum required sample size was 32 participants. A total of forty-eight semi-professional soccer players ($M_{age} = 22.4$, $SD_{age} = 2.25$) participated in this investigation. Players were members of three different teams (two male teams, $M_{age} = 22.9$, $SD_{age} = 5.60$; and one female team, $M_{age} = 21.9$, $SD_{age} = 6.20$), which performed three training sessions per week (90–100 min per session) and played a competitive

match every week (at least) in an official national league. All players have at least 10 years of experience playing soccer and have not reported previous injuries that limit their normal participation in the research. Nonsignificant differences between the male and female players were observed in the study. All players were warned not to take any caffeine or creatine before the research training sessions.

The procedures of this study were approved by the Ethics Local Committee (University of Extremadura; protocol number: 93/2020). Before starting this research, participants signed an informed consent that explained the purposes and potential risks associated with participating in this research. All data were treated according to the privacy, ethics, and protection policies of the American Psychological Association (2020).

2.2. Instruments

Polar Team Pro. To assess the physical load of the training tasks, we used the Polar Team Pro. This Global Position System is based on a signal concentration system of different Polar brand sensors, designed to control physical activity in collective sports (Coutts & Duffield, 2010). Specifically, we measured: mean Heart Rate (Mean HR) and maximum HR (Peak HR) as variables of internal load, and distance covered per minute (Distance/min.), mean speed, maximum speed, and number of sprints (Sprints) as variables of external load. This tracking technology has been validated (Huggins et al., 2020), and it is one of the most frequently used instruments for external load monitoring in soccer scenarios (Buchheit et al., 2018; García-Calvo et al., 2019, 2021).

Rating of Perceived Exertion. To assess the players' internal load through their perceptions of effort in the different tasks, the Rating of Perceived Exertion (RPE) was measured using the CR-10 scale. This instrument includes the following values: 0 (*Rest*), 1 (*Very, very easy*), 2 (*Easy*), 3 (*Moderate*), 4 (*Somewhat hard*), 5, and 6 (*Hard*), 7, 8, and 9 (*Very Hard*), and 10 (*Maximal*). The accuracy of this procedure for the purpose described has been proved in soccer tasks (Impellizzeri et al., 2004).

NASA-Task Load Index Questionnaire. To quantify the subjective feelings of mental load, we used an adaptation of the NASA – Task Load Index Questionnaire (NASA-TLX). The responses ranged from 0 (*no effort perceived*) to 100 (*maximum effort perceived*). In this adaptation, the players rated different soccer-related issues: mental effort, physical effort, time pressure, performance satisfaction, general effort, insecurity, and effort to maintain interaction with teammates. This instrument has been previously applied for this purpose in soccer (Díaz-García et al., 2021; García-Calvo et al., 2019, 2021).

Visual Analogue Scale. To quantify the subjective feelings of mental fatigue, the Visual Analogue Scale (VAS) was used. The responses ranged from 0 (*no fatigue perceived*) to 100 (*maximum fatigue perceived*). The accuracy of this scale to measure mental fatigue in soccer has been verified (Smith et al., 2017).

Before each session, all players put on the GPS Polar Team Pro to record physical load (Malone et al., 2017). After each task, VAS, the adaptation of NASA-TLX, and RPE were administered for each task separately during the rest periods. Players had previous experience with these methodological and assessment instruments, as they had undergone an initial measurement that was not considered for the study to ensure that they understood all these aspects.

2.3. Design

The research design was quasi-experimental. Two sessions were carried out with the same four training tasks: two possession games, where the goal was to perform a previously indicated number of consecutive passes, and two simulated matches with goalkeepers and traditional goals. To test the effect of the available time, in one session, there was more time available to solve the task, while in the other session, there was less time available. Specifically, to manipulate the

available time to solve the goal of the task, we used the score as a constraint (i.e., the initial or partial result of the task). In one session, teams started out losing (i.e., 2–0 or 0–2), and their task was to solve this situation. In the other session, the task began with the teams drawing (i.e., 0–0) and, after a previously determined time, the coaches and staff indicated that the score was 2–0 or 0–2. From then on, the teams had to finish the task to solve this situation (i.e., less available time than in the previously described situation). The total duration of the task was the same in both sessions. The order of the team that started with an unfavorable result (i.e., losing 2–0 goals) was counterbalanced to avoid the effects of learning on load. There was a recovery time of 3 min between Tasks 1 and 2, and 5 min between Tasks 3 and 4. All this information is included in Table 1.

Researchers aimed to control as many variables as possible to reduce the effect of variables other than the available time on the mental fatigue produced by these tasks. Therefore, coaches were instructed not to give feedback or encouragement during the tasks, and the players were grouped in the same way between sessions. All the sessions were video recorded to determine any event occurring during the different tasks. Furthermore, to control the possible effects of the contaminating variables, the sessions were supervised by a research assistant who registered all unexpected events, problems, or additional anecdotal information in a field notebook.

2.4. Data analysis

The statistical program SPSS 25.0 was used for data analysis. Firstly, to check data normality, a Shapiro-Wilk test was conducted. Also, a prior analysis of the data, using gender as a factor, was performed to avoid the possible influence of gender on the results. No differences were observed between male and female teams in the study variables. All results are presented as means (*M*) $\pm$ standard deviation (*SD*). To test the effects of the available time on physical load and mental fatigue, a *t*-test was carried out between all the variables, comparing the task with more available time (i.e., the entire time of the task to solve the situation) versus the same task with less available time (i.e., part of the time of the task to solve the situation). To further analyze this constraint according to the type of task, all this information was analyzed separately for possession games and simulated matches. Furthermore, effect size (*ES*) comparisons were conducted using Cohen's *d* (Cohen, 1988) and the magnitude-based inferences (MBI) with the following thresholds (Batterham & Hopkins, 2006): trivial (<0.2), small (0.2–0.6), moderate

(0.6–1.2), large (1.2–2.0), and very large (>2.0). To assess the likelihood of an effect being greater than the smallest important difference, the smallest worthwhile change (SWC) was calculated and classified as 0.5%–5%, very unlikely; 5%–25%, unlikely; 25%–75%, possibly; 75%–95%, likely; 95%–99.5%, very likely; and >99.5%, most likely (Hopkins et al., 2009).

3. Results

3.1. Mental load and fatigue

Table 2 shows the results of the mental load and fatigue produced by the tasks. Most of these variables showed higher values of mental effort and fatigue when the time pressure increased. Concretely, we found significantly higher values in mental ($p = .007$; $ES = 0.32$) and physical effort ($p = .009$; $ES = 0.39$) and time pressure ($p < .001$; $ES = 0.65$) when there was less available time in the possession task. In fact, the general effort was significantly higher in simulated matches in the condition of less available time ($p = .042$; $ES = 0.37$). This result suggests that the soccer players perceived the tasks with less available time as more mentally demanding.

3.2. Internal physical load

Table 3 shows the results of the internal physical load of the tasks in terms of Mean and Peak HR and RPE. In the possession games and simulated match tasks where there was more available time, HR (Mean, $p < .01$, $ES = -0.77$; and Peak, $p < .01$; $ES = -0.41$ for possession games; Mean, $p < .001$, $ES = -0.97$; and Peak, $p < .001$; $ES = -0.54$ for match games) was significantly higher according to the results showed in the *t*-test and MBI. Therefore, HR was higher under neutral conditions (i.e., more available time) than under time constraints (i.e., less available time). On the contrary, the RPE was higher in possession-games and simulated-match tasks with less available time ($p > .05$). Although no significant changes were observed in the *t*-test, a very likely + ive change was observed in RPE during the possession task when there was less available time. These results indicate that the players' subjective perceptions of physical exertion were higher both in possession games and simulated matches when the time pressure increased (i.e., less available time), although the physical efforts measured through HR were higher when time pressure decreased (i.e., more available time).

3.3. External physical load

Finally, Table 4 shows the data obtained in external physical load by distance/min, peak and mean speed, and number of sprints between tasks. Significantly higher distances/min ($p < .001$; $ES = -2.21$ for possession games; $p < .001$; $ES = -0.62$ for match games) and mean speeds ($p < .001$; $ES = -2.15$ for possession games; $p < .001$; $ES = -0.65$ for match games) were confirmed by *t*-test and MBI both in possession games and simulated matches when time pressure was lower. Peak speed was significantly higher in possession tasks with less time pressure ($p < .001$; $ES = -1.22$), with no significant changes between conditions in simulated matches. No significant differences in the number of sprints were observed between conditions in any of the tasks. These results suggest that the differences observed between conditions occur in aerobic variables such as distance/min and mean speed, with fewer changes in anaerobic variables such as peak speed and number of sprints.

4. Discussion

This study aimed to verify whether soccer players' available time to solve a task affects their mental load and fatigue as well as their physical training load. According to the main findings of the present study, soccer players reported that tasks with less available time were more mentally demanding. Moreover, when players had less available time to achieve

Table 1
Design of the intervention.

Task Order	Session with more available time	Session with less available time
1	Possession (4 vs 4). Goal is achieved every five passes. Field 30m $\times$ 20m. Five minutes long. One team starts losing 2–0 goals.	Possession (4 vs 4). Goal is achieved every five passes. Field 30m $\times$ 20m. Five minutes long. Score started 0–0. After 2 min, one team losing 2–0 goals, regardless of the real result.
2	Possession (4 vs 4). Goal is achieved every five passes. Field 30m $\times$ 20m. Five minutes long. The other team (counterbalanced order between different teams) starts losing 2–0 goals.	Possession (4 vs 4). Goal is achieved every five passes. Field 30m $\times$ 20m. Five minutes long. Score started 0–0. After 2 min, the other team losing 2–0 goals, regardless of the real result.
3	Match (8 vs. 8). Normal goal. Field 70 m $\times$ 40m. 12 min long. One team starts losing 1–0 goals.	Match (8 vs. 8). Normal goal. Field 70m $\times$ 40m. 12 min long. Score started 0–0. After 7 min, one team losing 1–0 goals, regardless of the real result.
4	Match (8 vs. 8). Normal goal. Field 70 m $\times$ 40m. 12 min long. The other team starts losing 1–0 goals.	Match (8 vs. 8). Normal goal. Field 70m $\times$ 40m. 12 min long. Score started 0–0. After 7 min, the other team losing 1–0 goals, regardless of the real result.

Table 2

A comparison of mental load and fatigue values according to time available by type of task.

Variables		Possession		Match	
		More available time	Less available time	More available time	Less available time
Mental Effort	M ± SD	60.49±20.55	69.10±18.59	68.31±17.83	72.98±14.56
	t(p)	−3.50(**)		−1.81(.145)	
	ES	.32		.19	
	% (+/trivial/-) QI	99/0/1 Very likely + ive		87/0/13Unclear	
Physical Effort	M ± SD	67.46±19.53	74.18±11.73	74.92±11.33	75.32±12.01
	t(p)	−2.86(**)		−.81(.419)	
	ES	.39		.11	
	% (+/trivial/-) QI	100/0/0Most likely + ive		80/0/20Unclear	
Temporal pressure	M ± SD	61.95±16.45	73.36±15.08	66.45±14.91	67.33±17.14
	t(p)	−4.75(***)		.13(.919)	
	ES	.65		−.09	
	% (+/trivial/-) QI	100/0/0 Most likely + ive		33/0/67Unclear	
Performance Satisfaction	M ± SD	66.69±16.12	66.56±18.43	68.00±16.08	71.94±16.20
	t(p)	.34(.937)		−1.64(*)	
	ES	.03		.18	
	% (+/trivial/-) QI	60/0/40Unclear		87/0/13Unclear	
General Effort	M ± SD	71.53±18.94	75.49±12.77	73.00±14.99	78.15±11.60
	t(p)	−1.72(.140)		−2.74(*)	
	ES	.27		.37	
	% (+/trivial/-) QI	98/0/2Unclear		100/0/0Most likely + ive	
Insecurity	M ± SD	42.29±24.32	44.34±22.56	46.67±24.44	47.26±24.47
	t(p)	−.83(.250)		−.34(.651)	
	ES	.05		.16	
	% (+/trivial/-) QI	63/0/37Unclear		83/0/17Unclear	
Interaction	M ± SD	66.95±17.15	70.49±20.57	68.50±17.45	68.31±22.87
	t(p)	−1.75(.059)		.07(.963)	
	ES	.06		−.09	
	% (+/trivial/-) QI	66/0/34Unclear		24/0/76 Unclear	
Mental Fatigue	M ± SD	57.29±24.88	60.74±17.95	58.58±22.63	59.92±18.37
	t(p)	−1.39(.178)		−.63(.601)	
	ES	.12		−.04	
	% (+/trivial/-) QI	82/0/18Unclear		39/0/61Unclear	

Note. * $p < .05$; ** $p < .01$; *** $p < .001$; ES = Effect Size; QI = Qualitative Inference; Without limitation: players have more time; With limitation: players have less time.

Table 3

A comparison of internal load values according to time available by type of task.

Variables		Possession		Match	
		More available time	Less available time	More available time	Less available time
Mean HR	M ± SD	169.48±12.77	160.95 ± 21.77	172.03 ± 10.48	162.28 ± 11.40
	t(p)	3.28(**)		8.58(***)	
	ES	−.77		−.97	
	% (+/trivial/-) QI	0/0/100Most likely -ive		0/0/100Most likely -ive	
	QI				
Peak HR	M ± SD	181.71 ± 11.94	177.21 ± 16.01	186.45 ± 9.52	181.51 ± 10.24
	t(p)	2.51(**)		5.42(***)	
	ES	−.41		−.54	
	% (+/trivial/-) QI	1/0/99Very likely -ive		0/0/100Most likely -ive	
	QI				
RPE	M ± SD	7.10 ± 1.84	7.57 ± 1.39	7.62 ± 1.22	7.79 ± 1.32
	t(p)	−1.83(.094)		−1.42(.197)	
	ES	.28		.12	
	% (+/trivial/-) QI	98/0/2Very likely + ive		84/0/16Unclear	
	QI				

Note. ** $p < .01$; *** $p < .001$; ES = Effect Size; QI = Qualitative Inference; Without limitation: players have more time; With limitation: players have less time.

the goals of the tasks, they significantly increased their RPE, and decreased their HR and the distances covered, without significant changes in anaerobic efforts.

We hypothesized that less available time would imply higher values of mental fatigue (*Hypothesis 1*). Although mental fatigue values were higher both in possession games and simulated matches when the available time decreased, no significant differences were observed between the conditions of available time. Therefore, *Hypothesis 1* can be confirmed. Soccer players found the task with less available time to be significantly more mentally demanding, based on the results obtained in the adaptation of NASA-TXL. As [Van Cutsem et al. \(2017\)](#) stated, cognition and emotions can cause mental demands in sports. In this case, the mental demands produced by an increase in the time pressure could lead to an increase in cognitive complexity and difficulty in controlling emotions (i.e., stress or frustration). Indeed, the intrinsic design of the tasks used, where available time was modified by a change in the match status, may provoke emotional states and motivational changes in the players ([Englert et al., 2015](#)). As explained previously, according to the ego-depletion theory, the constraints included in this design may cause a significant increase in mental fatigue and impair performance. In a previous study, [Sampaio et al. \(2014\)](#) stated that match status varied soccer players' collective behavior, and [Ferraz et al. \(2018\)](#) observed an increase in pacing when players had limited time to reverse the score in small-sided games. Therefore, the entropy of this change and the difficulty of these new collective behaviors (i.e., cognitive complexity) may increase players' reported mental load. Although no significant differences were observed, players reported higher insecurity and interaction when time pressure increased, suggesting a potential role of emotions in the results described. However, although mental load and mental fatigue are typically associated, many factors may also influence athletes'

Table 4

A comparison of external load values according to time available by type of task.

Variables		Possession		Match	
		More available time	Less available time	More available time	Less available time
Distance/ min.	M ± SD	97.55 ± 8.33	82.15 ± 13.92	101.95 ± 11.05	95.38 ± 11.15
	t(p)	7.43(***)		5.02(***)	
	ES	−2.21		−.62	
	%	0/0/100	Most likely -ive	0/0/100	Most likely -ive
	(+/trivial/-) QI				
Peak Speed	M ± SD	19.61 ± 2.46	16.77 ± 2.46	23.36 ± 2.76	23.37 ± 2.74
	t(p)	5.86(***)		−.17(.457)	
	ES	−1.22		.00	
	%	0/0/100	Most likely -ive	51/0/49	Unclear
	(+/trivial/-) QI				
Mean Speed	M ± SD	6.22 ± .50	5.31 ± .89	6.39 ± .64	6.00 ± .68
	t(p)	7.30(***)		4.90(***)	
	ES	−2.15		−.65	
	%	0/0/100	Most likely -ive	0/0/100	Most likely -ive
	(+/trivial/-) QI				
Sprints	M ± SD	.73 ± 1.20	.40 ± .90	1.68 ± 1.87	1.95 ± 1.65
	t(p)	2.15(*)		−1.11(.215)	
	ES	.07		.19	
	%	56/0/44	Unclear	88/0/12	Unclear
	(+/trivial/-) QI				

Note. * $p < .05$; *** $p < .001$; ES = Effect Size; QI = Qualitative Inference; Without limitation: players have more time; With limitation: players have less time.

mental fatigue. Van Cutsem and Marcora (2021) found that mental fatigue could depend on task difficulty, engagement, task enjoyment/aversion, or task duration. Although these two sessions used the same tasks, the constraint could modify any of these variables, producing smaller increases in mental fatigue than the desired outcomes according to the mental load reported. Therefore, these results suggest that time pressure significantly increased the reported mental load, which may be mediated by increased cognitive complexity and emotions. However, the mental fatigue increases associated with this constraint may be reduced by other related factors such as engagement (Van Cutsem & Marcora, 2021).

We also hypothesized that less available time would lead to higher values of internal physical load (*Hypothesis 2a*). On the contrary, mean and peak HR showed significantly higher values in tasks with more available time, whereas RPE was higher in tasks with less available time. Therefore, *Hypothesis 2a* could not be confirmed. It has been suggested that a negative match status may increase losing teams' physical efforts (Castellano et al., 2011; Sampaio et al., 2014). However, it has also been shown that knowing the length of the task may decrease players' physical efforts because they develop a profile of efforts to ensure their performance across the entire task (Ferraz et al., 2018). This information about task duration may influence the results obtained. However, the results in RPE suggest a negative influence of mental load and fatigue on physical efforts. Specifically, effort perceptions were higher when there was less available time, although internal physical variables (i.e., mean and peak HR) were higher when there was more available time. Van Cutsem et al. (2017) indicated that RPE increases mediate the negative influence of mental fatigue on sports physical performance. According to Marcora et al. (2009), these significantly higher values of mental fatigue may increase the perceptions of RPE, impair self-control, and decrease players' physical efforts when the available time decreases. Under time constraints, mental fatigue may increase RPE and, consequently, decrease players' physical efforts (e.g., the smallest HR observed in the

experimental condition). Therefore, knowledge about how the available time and the mental load and fatigue caused by this constraint affect players' physical performance and feelings of physical effort is essential.

Finally, we also hypothesized that less available time would imply higher values of external physical load (*Hypothesis 2b*). On the contrary, distance/min and mean speed decreased significantly when there was less available time to solve the task, without changes in peak speed (in simulated matches) or sprints (in both tasks). Although in the previous paragraph, we associated these changes with collective behaviors and knowledge about task duration, these results reinforce the idea of the influence of mental fatigue on physical efforts, because it has been shown that mental fatigue influences aerobic performance without changing anaerobic performance (Habay et al., 2021). Therefore, the mental load and fatigue caused by this constraint may reduce players' physical effort during these tasks.

5. Conclusions

Based on the results of the present study, we can conclude that increasing the time pressure through the scoring system and the task duration significantly increases players' reported mental load. Although mental fatigue was higher when there was more time pressure, the changes observed in this variable were nonsignificant. The increased mental load and fatigue caused by this constraint influence physical performance, as soccer players also reported higher feelings of RPE, showing a relationship between mental fatigue and RPE. However, tasks with time pressure implied a decrease in external physical load and aerobic effort values.

5.1. Limitations, future research, and practical applications

Some limitations should be commented on, requiring new investigations to achieve greater empirical strength. One of the limitations was the small sample size and the low representativeness of the constraints used, as they were only measured in two sessions. More research is needed, with more teams, in different contexts, and with a larger number of sessions, as well as different types of soccer tasks. It is, therefore, necessary to continue investigating mental and physical load and mental fatigue in new training situations to better understand these two constructs and to design and plan training adequately. Furthermore, researchers should consider that this type of ecological design may include many variables that influence mental fatigue (coaches, teammates, etc.). Therefore, this limitation is present in most of these researches. For this purpose, the present study used the VAS-100 scale, whereas, in the future, other instruments may add more information about soccer players' mental fatigue. Finally, we did not analyze physical fatigue, and this construct is highly related to physical load (Hadi et al., 2020), so it would be interesting to consider it in new studies. Furthermore, we did not consider other non-sport burden activities (e.g., sleep, stress, etc.), which should be considered in future studies.

Several practical contributions can be extracted from the findings obtained in this study: the results obtained could be considered when designing training tasks that simulate situations that may occur during competitions to generate players' adaptations. We underline the importance of controlling and quantifying the training load and its corresponding fatigue through the strategy proposed to coaches and technical staff. Soccer is both mentally and physically fatiguing, so coaches should plan training sessions with mental load doses to develop players' adaptation to mental fatigue in soccer situations. On the other hand, to optimize the training process, coaches should consider different levels of mental load to be employed in soccer tasks according to the training day of the soccer micro-cycle or season stage. Finally, understanding the behavior of training load and mental fatigue in players allows for correctly designing and planning the training process, controlling the load to optimize players' performance.

